

Borrowing Strength Across Vulnerabilities: Hierarchical and Type-Aware Forecasting of Sparse Sightings

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Abstract

Forecasting how often a vulnerability will be *seen*—in proof-of-concept releases, scanner detections, exploitation reports, or online discussion—is a practical aid to defenders, but the data are notoriously hard: per CVE the series are short (often 10–30 days), mostly zero, and punctuated by bursts. Our earlier work [3] established *qualitatively* that classical time-series models (SARIMAX) fail in this regime and that count-based and decay models behave more sensibly, but it could neither rank the surviving models nor quantify the gains of any proposed improvement. This paper supplies the missing quantitative foundation and a concrete improvement. First, we build a reproducible evaluation harness based on proper scoring rules (CRPS, quantile loss, RMSSE) and rolling-origin backtesting, and use it to show that a hurdle Negative-Binomial model—which explicitly handles the over-dispersion and excess zeros the first paper identified—is the best-calibrated single-series model, while a time trend is actively harmful. Second, and most importantly, we attack the root cause—per-CVE data scarcity—with *hierarchical (partial) pooling*: each CVE’s activity and burst parameters are shrunk toward a population distribution estimated from a back-catalogue of past CVEs. This dominates both the strong rolling-mean baseline and the best unpooled model at every training-window size and yields roughly a threefold improvement in data efficiency. Finally, we model sighting *type*: pooling cleanly forecasts the operationally high-value *exploited* signal, although decomposing the total by type does not help the aggregate, and—contrary to intuition—open-source proof-of-concept and “seen” activity *co-move* with exploitation rather than reliably leading it. All code, data snapshots, tables, and figures are released for reproducibility.

Topics: *Measuring vulnerabilities, Forecast, Decision*

1 Introduction

A vulnerability *sighting* is a discrete, time-stamped observation that a specific CVE is being referenced, discussed, or exploited—a proof-of-concept (PoC) publication, a scanner detection template, an entry in an exploited-in-the-wild catalogue, or a post across the Fediverse [1]. The number of sightings a vulnerability accrues over time is a direct, observable proxy for the attention and activity it attracts, and forecasting it—even coarsely—would help defenders anticipate which vulnerabilities are about to matter.

In a previous study [3] we examined the feasibility of this forecasting task and reached a largely cautionary conclusion. Sighting series are *sparse* (mostly zeros), *bursty* (a PoC release or a wave of scans produces a sudden spike), and *short* (a CVE is typically only a few weeks old when a forecast would be useful). Under these constraints, seasonal SARIMAX models produced unstable, sometimes negative forecasts with confidence intervals spanning orders of magnitude; count-based models (Poisson regression) and simple decay curves were more sensible but still fragile. Crucially, that study assessed models *by eye*, from plots. This is sufficient to reject SARIMAX, but it cannot answer the questions that matter for a deployment: *Which* of the surviving models is best? *By how much*? And does a proposed refinement genuinely help, or merely look plausible on a hand-picked example?

This paper provides the quantitative answers and a substantive methodological advance. Our contributions are:

1. **A reproducible probabilistic evaluation harness** (Section 4) for sparse sighting forecasts, combining rolling-origin backtesting with strictly proper scoring rules (CRPS, quantile loss, RMSSE) and calibration diagnostics (interval coverage, randomised PIT). This turns the first paper’s qualitative claims into measurements.
2. **Count models matched to the data’s pathologies** (Sections 5 and 6). We show that a *Negative-Binomial* model fixes the over-dispersion the first paper observed, and that a two-part *hurdle* addresses its excess zeros; the hurdle Negative-Binomial is the best-calibrated single-series model. We also show that fitting a time *trend*—the form used previously—is actively harmful, reproducing the SARIMAX blow-up in count form.
3. **Hierarchical pooling across CVEs** (Section 7), the central result. Rather than fitting each CVE in isolation, we treat its parameters as draws from a population distribution learned from past CVEs and shrink accordingly. This dominates every alternative at all training-window sizes and gives roughly a $3\times$ data-efficiency gain—directly addressing the scarcity that defeated the first paper.
4. **Type-aware forecasting** (Section 8), realising a stated goal of the first paper. Pooling the *exploited* stream forecasts that high-value signal well; however, decomposing the total by type does not improve the aggregate, and a lead-lag analysis shows that PoC/“seen” activity does not cleanly precede exploitation in this data—a cautionary, actionable finding.

The implementation, the cached sighting corpus, and the scripts that regenerate every table and figure in this paper are released as open source.

2 Background and Related Work

Vulnerability activity forecasting. Most vulnerability prioritisation relies on *assessment* rather than *prediction*: CVSS severity, or our own VLAI severity model [2], which predicts CVSS-like scores from descriptions. The Exploit Prediction Scoring System [10] predicts the *probability* of exploitation, and Vuln4Cast [11] forecasts vulnerability *publication* volumes at an aggregate level. Our task differs: we forecast the *temporal count* of observable activity for an individual CVE. The direct predecessor of this work is [3], whose diagnosis (sparse, bursty, short series; SARIMAX unsuitable; count and decay models preferable) we take as our starting point.

Count and intermittent-demand models. Non-negative integer outcomes are naturally modelled by Poisson or Negative-Binomial regression; the latter adds a dispersion parameter for the common case where variance exceeds the mean. When data are dominated by zeros, hurdle models [13] separate the occurrence of any activity from its magnitude. The pattern of long zero runs with occasional positive counts is precisely *intermittent demand*, for which Croston’s method [4] and the Syntetos–Boylan approximation [16] are the field-standard estimators; we use them as baselines.

Borrowing strength. Estimating a parameter for a data-poor unit by shrinking it toward a population estimated from many related units is a classical idea—Stein estimation and empirical Bayes [7], and modern multilevel modelling [8]. Global/cross-learning forecasters such as DeepAR [14] exploit the same intuition at scale. We bring this lever to vulnerability sightings, where per-unit scarcity is the dominant obstacle.

Forecast evaluation. A probabilistic forecast should be assessed with a strictly proper scoring rule [9]; the continuous ranked probability score (CRPS) is the natural choice and, for count data, calibration is checked with the randomised probability integral transform [6]. Scaled point errors (RMSSE) as popularised by the M5 competition [12] allow aggregation across series of very different magnitudes. Our harness combines these.

3 Data and Problem Setup

We obtain sightings from the public Vulnerability-Lookup API operated by CIRCL. Each record carries a creation timestamp, a `type` (see Section 8), and a source. For a given CVE we aggregate records into a gap-free daily count series, filling days with no sightings with explicit zeros so that the calendar is regular. The forecasting target is the next H daily counts ($H = 7$ throughout).

Corpus. To study both the data-scarce and data-rich regimes, and to provide a credible *population* for pooling, we assemble 24 CVEs: the cases analysed in [3] plus 18 high-profile exploited vulnerabilities (e.g. Log4Shell CVE-2021-44228, ProxyLogon CVE-2021-26855, MOVEit CVE-2023-34362, Cisco IOS XE CVE-2023-20198). Histories range from a single 11-day series (CVE-2024-9164) to nearly four years (CVE-2022-26134). Figure 1 illustrates the regime: a rapid burst after disclosure, a heavy-tailed fade, and long stretches of zeros—exactly the structure that frustrates smooth time-series models.

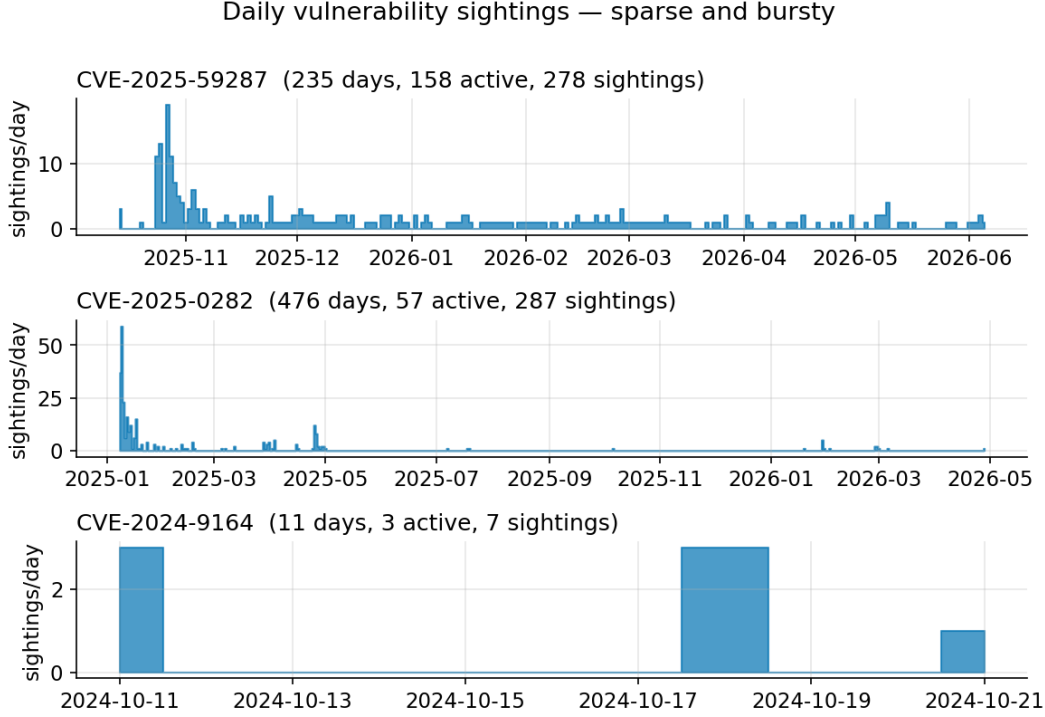


Figure 1: Daily sighting counts for three CVEs spanning the regime: a bursty, fading series (top), a sharp spike followed by near-silence (middle), and an extremely short 11-day series (bottom). Counts are low, spiky, and dominated by zeros.

4 Evaluation Methodology

The first paper’s reliance on visual inspection is the gap we close first. Three design choices make model comparison fair and quantitative.

Sample-based probabilistic forecasts. The models we compare—count GLMs, a two-part hurdle, intermittent-demand baselines, and the hierarchical model of Section 7—share no common closed-form predictive distribution, but each can *simulate* futures. We therefore represent every forecast as a pool of S Monte-Carlo sample paths of length H , and compute all metrics from those samples. This single representation lets one scoring routine treat every model identically.

Rolling-origin backtesting. Because series are short, a single train/test split would confound model quality with the accident of where the split falls. We instead slide the forecast origin forward one day at a time: at origin t the model is fit on observations $1:t$ and scored on $t+1:t+H$. Only past data is ever used, and each CVE yields many scored forecasts, so comparisons rest on a distribution of errors.

Scoring rules. For predictive samples $\{x_h^{(s)}\}_{s=1}^S$ and realised count y_h at horizon step h , we report:

- **CRPS** [9], our primary metric, estimated from samples as

$$\text{CRPS} = \frac{1}{S} \sum_s |x^{(s)} - y| - \frac{1}{2S^2} \sum_{s, s'} |x^{(s)} - x^{(s')}|, \quad (1)$$

which generalises absolute error to distributions and is in the units of the data (sightings/day);

- **quantile (pinball) loss** averaged over a grid of levels;
- **RMSSE** [12], the point error scaled by the in-sample one-step naive error, which is unit-free and comparable across CVEs;
- **interval coverage** at the 50/80/95% levels; and
- **randomised PIT calibration error** [6]: the mean absolute deviation of the PIT histogram from uniform, our reliable calibration gauge (raw coverage is inflated by structural zeros, see below).

5 Count Models for Sightings

Baselines. A complex model must beat cheap ones to justify itself, and RMSSE is *defined* relative to a naive forecast. We include *persistence* (last value), a *7-day rolling mean*, and Croston’s method with the Syntetos–Boylan correction [4, 16]. Each turns a point rate into Poisson samples so it is a genuine probabilistic forecast.

Poisson and Negative Binomial. The first paper settled on Poisson regression, which guarantees non-negative integer forecasts but imposes $\text{Var} = \text{mean}$. It explicitly reported observing both over- and under-dispersion, violating that assumption. The Negative-Binomial (NB2) model relaxes it with a dispersion parameter α ,

$$\text{Var}(y) = \mu + \alpha \mu^2, \quad (2)$$

collapsing to Poisson as $\alpha \rightarrow 0$, so it can only help at the cost of one parameter.

Hurdle. Sightings are mostly zeros with occasional bursts, and the process that decides *whether* a day is active may differ from the one that sets *how large* a burst is. A hurdle model [13] makes this explicit: a Bernoulli component for $\Pr(y > 0)$ and a zero-truncated count component for the magnitude given activity. This directly targets the excess zeros a single count model smears over.

Each count model is offered in a constant-rate (`const`) and a log-linear time-trend (`trend`) variant, the latter matching the `sightings ~ time` form used in [3], so the backtest can decide whether the trend earns its keep.

6 Experiment 1: Which Single-Series Model?

We backtest all models on five CVEs with sufficient history (origin-weighted, 2296 forecasts per model). Table 1 and Figure 2 report the result.

Table 1: Single-series model comparison (rolling-origin backtest, $H = 7$). CRPS and RMSSE: lower is better. PIT error: lower is better-calibrated.

model	CRPS	RMSSE	PIT cal. err.
rolling_mean_7	0.274	0.298	0.146
hurdle_negbin	0.308	0.362	0.071
hurdle_poisson	0.321	0.372	0.105
naive_last	0.321	0.342	0.172
negbin_const	0.377	0.541	0.230
poisson_trend	0.426	0.456	0.180
sba	0.564	0.461	0.258
poisson_const	0.579	0.618	0.282
negbin_trend	4.532	7.641	0.286

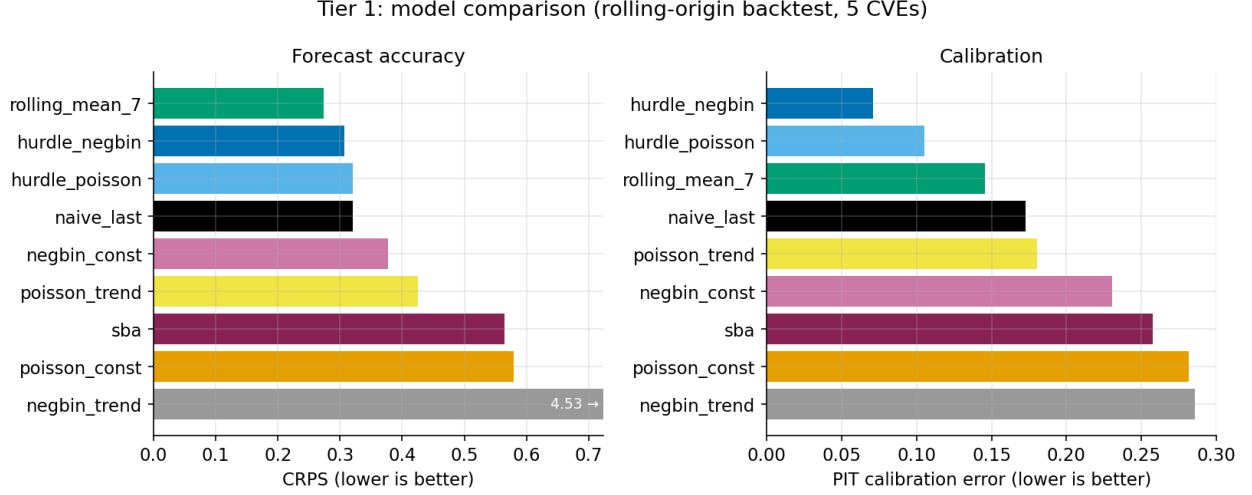


Figure 2: Forecast accuracy (CRPS) and calibration (PIT error) per model. The runaway `negbin_trend` is clipped off-axis and annotated.

Four findings stand out. **(i) Simple baselines are hard to beat:** the 7-day rolling mean has the lowest CRPS, confirming—now by measurement—the first paper’s qualitative remark that simple averages outperform elaborate time-series models here. **(ii) Negative Binomial beats Poisson** at every level, validating the over-dispersion fix. **(iii) The hurdle Negative-Binomial is the best-calibrated model** by a wide margin (PIT error 0.071 versus 0.282 for plain Poisson). When trustworthy prediction intervals matter more than a marginally sharper median—the operational case—it is the model of choice. **(iv) The time trend is a liability:** `negbin_trend` is the worst model overall, because exponential extrapolation of a fitted slope reproduces the SARIMAX blow-up in count form. Constant-rate count models are markedly more robust. We also note that raw interval coverage over-states calibration on these series (when the rate is low the central quantiles are both zero and the frequent zero actuals fall inside trivially); the PIT error is the gauge we rely on.

7 Hierarchical Pooling Across CVEs

Experiment 1 improves the *form* of the single-series model, but every such model is still starved of data: with 10–30 days per CVE the activity rate and burst size are estimated poorly. We now change the unit of learning.

7.1 Model

We keep the winning hurdle decomposition and assume each CVE c ’s parameters are draws from a shared population:

$$p_c \sim \text{Beta}(a, b), \quad \lambda_c \sim \text{Gamma}(\kappa, \theta), \quad (3)$$

where p_c is the probability a day is active and λ_c the mean burst size given activity. The population hyperparameters are estimated once from a back-catalogue of past CVEs (empirical Bayes, by method of moments), and each CVE’s estimate is the conjugate posterior mean, which shrinks toward the population in proportion to how little data the CVE has:

$$\hat{p}_c = \frac{k_c + a}{n_c + a + b}, \quad \hat{\lambda}_c = \frac{S_c + \kappa}{A_c + \theta}, \quad (4)$$

with k_c active days out of n_c , and S_c total sightings over A_c active days. Burst sizes are sampled as $1 + \text{NB}(\hat{\lambda}_c - 1)$ with a pooled dispersion, so they are strictly positive (the hurdle assumption). A short series starts near the population average and moves toward its own behaviour as evidence accrues—leverage a single-series model structurally lacks. We deliberately use empirical Bayes rather than full MCMC: it is dependency-light and instant, and suffices to establish whether pooling helps before investing in a fully Bayesian treatment (Section 10).

7.2 Experiment: data-starvation backtest

To isolate the effect pooling should have—helping when data is scarce—we restrict every model to a fixed trailing window of W days and sweep W from 5 to 30. The population prior is estimated *leave-one-out*: when forecasting CVE c it uses only the *other* CVEs’ histories, mirroring the realistic setting in which a new CVE arrives against an existing back-catalogue. We compare the hierarchical hurdle against the best unpooled model from Experiment 1 (`indep_hurdle_nb`) and the rolling-mean baseline.

7.3 Results

Figure 3 and Table 2 show that the hierarchical model has the lowest CRPS at *every* window size, with the largest margin where data is scarcest: at $W = 5$ it is $\approx 25\%$ better than the unpooled hurdle and $\approx 31\%$ better than the rolling mean. The population prior itself is informative—mean activity 0.21, mean burst rate 2.14, pooled dispersion $\alpha = 4.52$ (strong over-dispersion, a population-level justification for NB over Poisson).

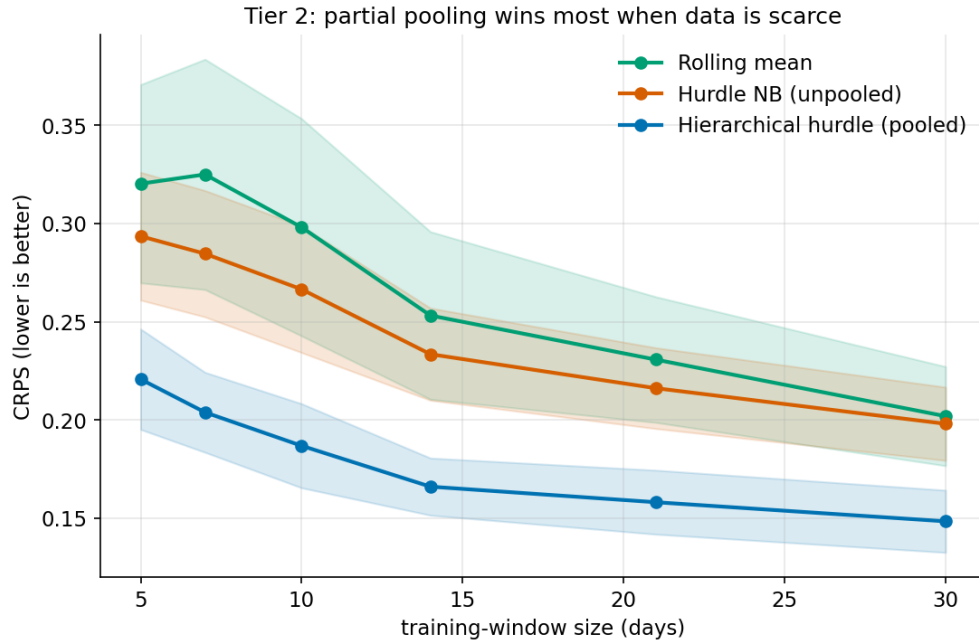


Figure 3: CRPS versus training-window size (mean $\pm$ SEM). The pooled hierarchical hurdle dominates the best unpooled model and the rolling-mean baseline at every window, most where data is scarcest.

Table 2: CRPS by training-window size W (days), 24-CVE data-starvation backtest. Lower is better.

model	$W=5$	7	10	14	21	30
hier_hurdle (pooled)	0.221	0.204	0.187	0.166	0.158	0.148
indep_hurdle_nb (unpooled)	0.293	0.285	0.266	0.233	0.216	0.198
rolling_mean	0.320	0.325	0.298	0.253	0.231	0.202

A useful way to read Table 2 is as **data efficiency**: the pooled model with only 10 days of history (0.187) already beats the unpooled hurdle with 30 days (0.198), and at 7 days it matches the unpooled model at 30 days. Pooling reaches in roughly a week what an isolated model does not reach in a month. Figure 4 makes the mechanism concrete on a single forecast: trained on a window ending in a burst, the pooled model shrinks its expectation toward the population and anticipates the fade, whereas the unpooled hurdle stays high and opens a very wide interval—the count-model echo of the first paper’s exploding confidence intervals.

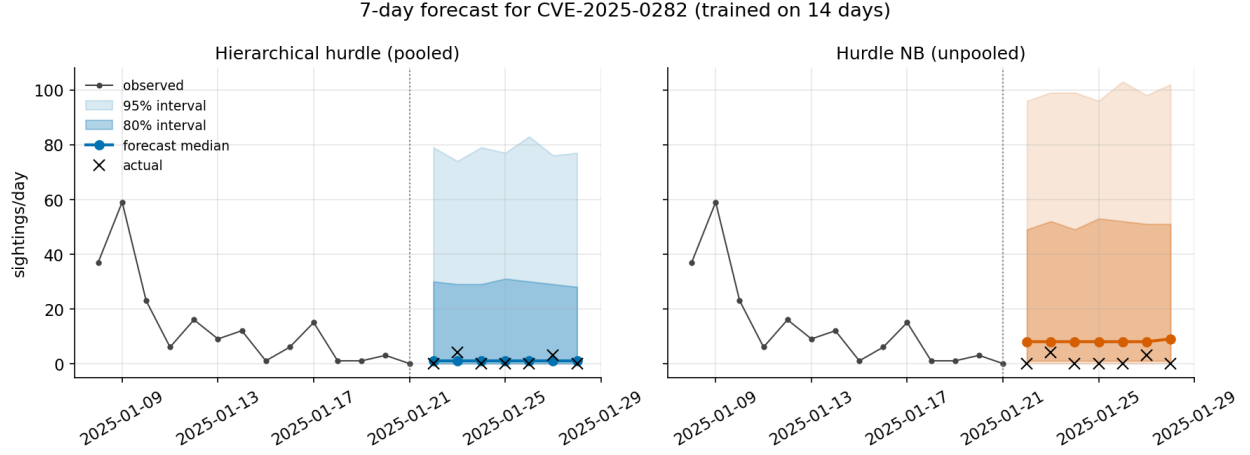


Figure 4: A 7-day probabilistic forecast (median, 80% and 95% intervals) against realised counts, trained on a 14-day window ending in a burst. Pooled (left) versus unpooled (right).

8 Type-Aware Forecasting

The first paper noted that treating all sightings as equivalent discards information, and proposed modelling their *type* to link activity to actual exploitation. Records carry three meaningful types—*seen* (background references), *published-proof-of-concept* (PoC), and *exploited* (exploitation reports, largely from the CISA KEV catalogue [5]); two further types are negligible ($<0.1\%$) and dropped. We pool *each type separately*, fitting one hierarchical hurdle per type with its own population prior, and form the total as the sum of per-type forecasts.

Types have distinct dynamics. Figure 5 and the per-type priors confirm the types are not interchangeable: PoC is rare and punctual (active on $\approx 2\%$ of days, low dispersion—a PoC drops once), *exploited* is the burstiest stream (dispersion $\alpha = 5.21$), and *seen* is steadier background chatter. This heterogeneity is precisely what a single pooled total blends away.

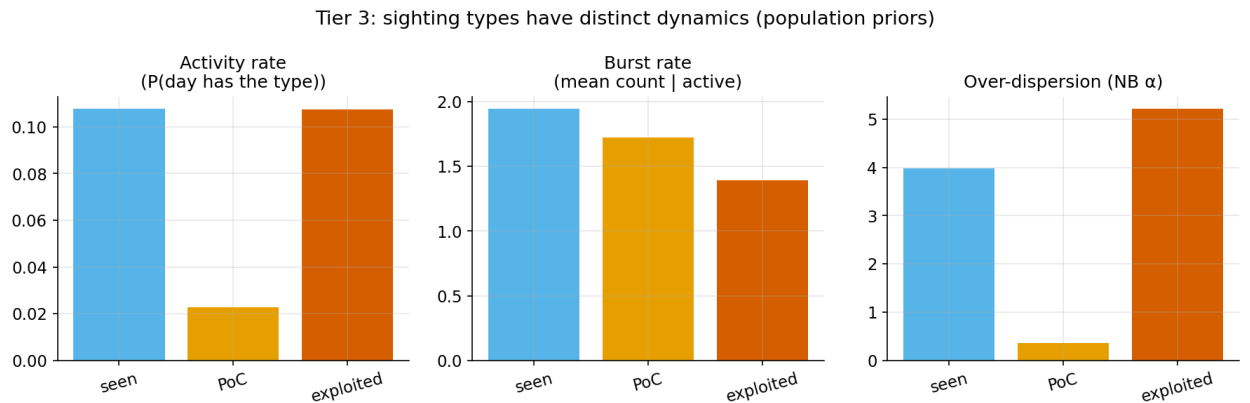


Figure 5: Population priors per sighting type: activity rate, mean burst size, and over-dispersion. The types differ markedly.

Pooling forecasts the exploited signal well. The per-CVE *exploited* stream is even scarcer than the total, so the pooling argument should apply with more force. Repeating the data-starvation backtest on the exploited-only series (Figure 6, Table 3), the type-specific pooled model again wins at every window by $\approx 25\text{--}30\%$ CRPS. We can thus forecast a CVE’s exploitation-related activity, with calibrated intervals, from very little of its own history—the operationally most valuable output.

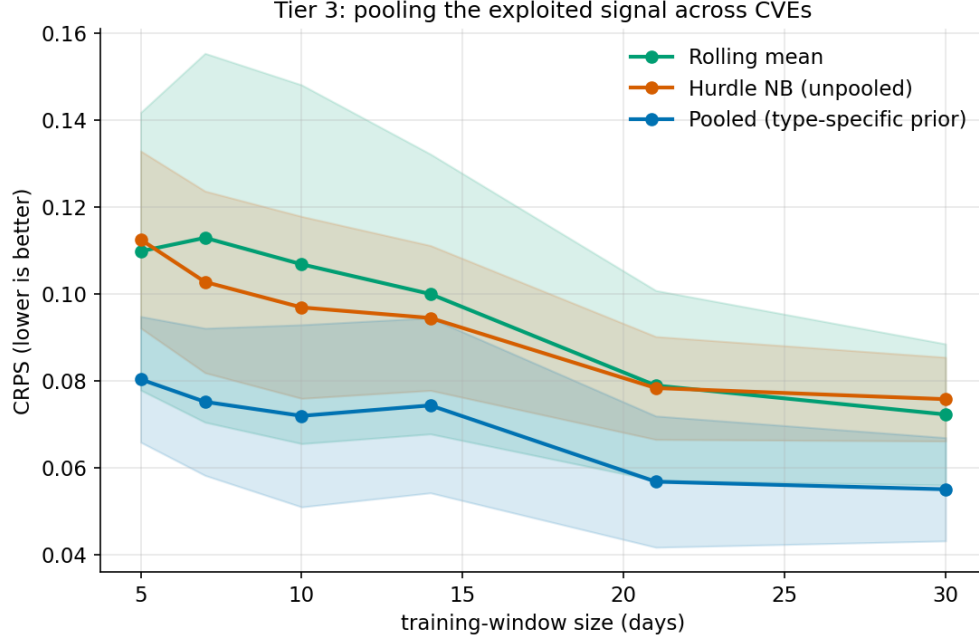


Figure 6: CRPS on the `exploited-only` series versus training-window size. Pooling the exploited signal across CVEs beats the unpooled hurdle and the rolling-mean baseline throughout.

Table 3: Exploited-signal CRPS by window W . Lower is better.

model	$W=5$	7	10	14	21	30
hier_exploited (pooled)	0.080	0.075	0.072	0.074	0.057	0.055
indep_hurdle_nb (unpooled)	0.112	0.103	0.097	0.094	0.078	0.076
rolling_mean	0.110	0.113	0.107	0.100	0.079	0.072

But decomposition does not help the total. A natural hope is that modelling types separately and summing improves the aggregate forecast. It does not (Table 4): the single pooled total is consistently, if slightly, better. Splitting the data across three components—and summing independent per-type samples, which inflates variance—costs more than the per-type priors buy back. The practical rule is therefore: *use the single pooled model for the total, and the typed model when a specific (e.g. exploited) forecast is wanted.*

Table 4: Total-count CRPS: typed decomposition versus single pooled model.

model	$W=7$	14	30
pooled_total (single)	0.203	0.165	0.148
typed_sum (decomposition)	0.207	0.177	0.155

Do PoC/seen sightings precede exploitation? The intuitive kill-chain suggests PoC publication should *lead* exploitation. We test this with the mean cross-correlation between each precursor type and `exploited`, over lags -14 to $+14$ days (positive lag = precursor leads), with each CVE’s series z-scored (Figure 7). The correlation *peaks at lag 0* (seen 0.32, PoC 0.29) and is roughly symmetric; if anything the negative lags (exploitation leading) are slightly stronger for PoC. In this data, open-source PoC/seen activity is therefore *not* a clean leading indicator of exploitation—the dominant relationship is contemporaneous. We attribute this largely to how `exploited` sightings are timestamped (KEV entries are added in clusters that need not respect the real PoC→exploitation ordering), and

read it as a caution against assuming open-source chatter leads exploitation. It motivates an event-time, first-occurrence analysis rather than daily counts (Section 10).

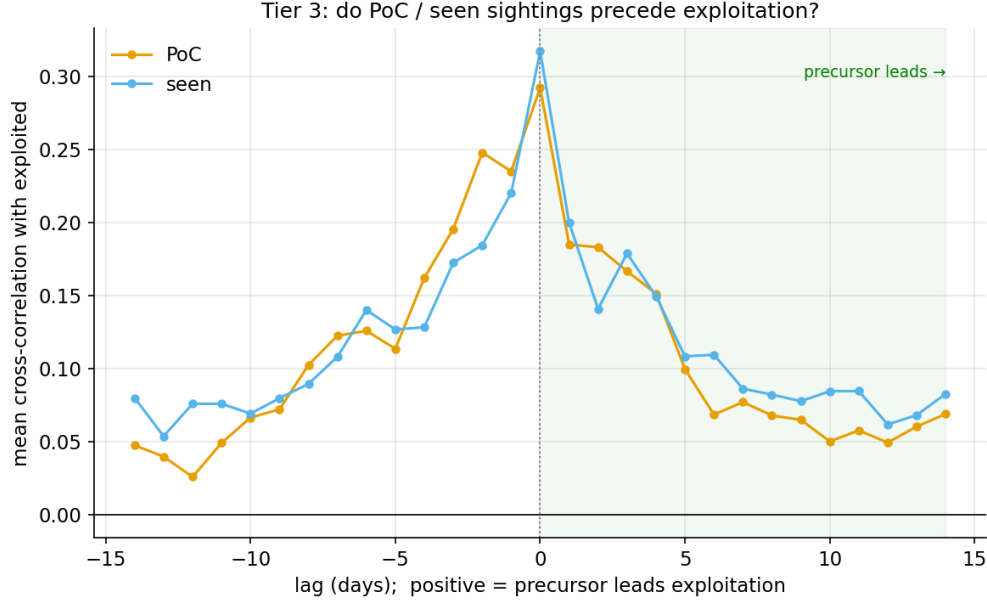


Figure 7: Mean cross-correlation of precursor types (PoC, seen) with `exploited` activity by lag. Positive lag means the precursor leads. The relationship peaks at lag 0 and is roughly symmetric—co-movement, not a clean lead.

9 Discussion: Operational Recommendations

The results translate into concrete guidance for a deployment such as Vulnerability-Lookup. (1) Adopt a probabilistic, count-based forecaster and report intervals, not point estimates; the *hurdle Negative-Binomial* gives the best calibration among single-series models. (2) Do not fit a time trend on short series—use constant-rate components. (3) When historical data is thin—the usual case for a freshly disclosed CVE—*pool across past CVEs*: a population prior is the single most effective lever we found, worth roughly a threefold increase in effective data. (4) For exploitation-relevant alerting, forecast the `exploited` type directly with its own pooled model rather than reading it off an all-types total. (5) Treat raw interval coverage with suspicion on sparse counts and rely on PIT-based calibration.

10 Limitations and Future Work

Our hierarchical model is an empirical-Bayes prototype: it uses point hyperparameters and so does not propagate hyperparameter uncertainty, it shares a single pooled burst dispersion rather than a per-CVE random effect, and it pools the activity and burst components independently. A fully Bayesian multilevel formulation [8] would address all three and likely close the small residual mis-calibration at the shortest windows. In the typed model the per-type streams are sampled independently; modelling their correlation (PoC and exploitation do co-occur) is a natural next step that could improve the aggregate. The lead-lag analysis uses daily counts; a point-process treatment on raw timestamps would test precedence more rigorously. Finally, time-varying covariates with genuine variation—day-of-week effects, and EPSS [10] dynamics rather than the near-constant severity score—and global cross-learning forecasters [14] are promising directions our harness is now equipped to evaluate.

11 Conclusion

The first study in this line concluded, qualitatively, that forecasting sparse and bursty vulnerability sightings is hard and that classical time-series models are ill-suited. This paper turns that diagnosis into measurement and offers a remedy. With a proper-scoring evaluation harness we show that a hurdle Negative-Binomial is the best-calibrated single-series

model and that time trends are harmful. More importantly, we show that the core obstacle—per-CVE data scarcity—is best attacked not by a cleverer single-series model but by *borrowing strength across vulnerabilities*: a hierarchical model that shrinks each CVE toward a population of past CVEs dominates all alternatives and roughly triples data efficiency, including for the high-value exploited signal. Along the way we report two findings that temper intuition: decomposing sightings by type does not improve the aggregate forecast, and open-source proof-of-concept activity does not cleanly precede exploitation in this data. We release all code, data, and figures to support reproducibility and further work.

Reproducibility

The implementation (data access, models built on statsmodels [15], the evaluation harness, and the scripts that regenerate every table and figure here) is available in the TARDISsight repository: <https://github.com/vulnerability-lookup/TARDISsight>.

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